

ROOPANSHU GUPTA

Software engineer and ML researcher building LLM agents and the benchmarks that grade them
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EDUCATION

Delhi Technological University — B.Tech in Software Engineering Expected May 2028
CGPA: 9.67 / 10 (Dept. Rank 1, Sem 2) | JEE Main AIR 7997 | ISI AIR 104 | Class XII 93.8% | Class X 97.33%

EXPERIENCE

Machine Learning Research Intern | Machine Learning Research Lab, DTU Jan 2026 – Present

- Ran an **ablation-driven research loop** on CycleGAN and conditional diffusion models for SAR-to-optical satellite image translation, raising **PSNR from 15 to 31.58 dB** and **SSIM from 0.18 to 0.87** via custom losses and attention.
- Engineered an end-to-end pipeline for **1TB+ satellite imagery** (acquisition, pairing, preprocessing, storage), eliminating manual dataset pairing for every experiment.
- Removed multi-GPU HPC training I/O bottlenecks with **AWS S3**-backed storage and batch prefetching across concurrent jobs; mentored junior students on deep-learning workflows.

5G AI Systems Intern | DoT 5G Use Case Lab, DTU Jun 2025 – Jul 2025

- Fine-tuned a **YOLOv8** traffic-analytics model to **97.5% mAP@0.50** and 90.0% precision; applied **FP16 quantization** to cut inference latency for real-time deployment on 5G edge hardware.

PROJECTS

Cloud Remediation Agent + RemediationBench | [GitHub](#)

Python, Claude API (tool use), Model Context Protocol, boto3, moto/LocalStack, pytest

- Built an incident-remediation agent with a **code-enforced detect → diagnose → policy-check → remediate → verify** loop: a deterministic policy gate checks every write, and a ticket cannot close until a probe confirms the reported symptom is gone.
- Built **RemediationBench**: 7 sandboxed AWS incidents (incl. red-herring decoys) graded by **hidden state verifiers** with collateral-damage checks. With a fixed-prior planner, the verify stage cut **false-success from 29% to 0%** (pass@1 71% → 100%); the policy gate blocked 2/2 unsafe shortcuts, while verifiers exposed **4 damaging actions no static rule caught**.

Unified Data MCP Server | [GitHub](#)

Python, MCP (Streamable HTTP), sqlglot, JWT / OAuth 2.1, SQLite / PostgreSQL / Snowflake, pytest

- Remote MCP server giving agents read-only access to SQL sources and allow-listed REST APIs via 5 typed tools and on-demand schema resources: **784 tokens** of tool context vs 1,368 for a one-tool-per-table design over 14 tables.
- Enforced JWT resource-server auth, role-based access, per-user rate limits and hashed audit logs; an AST-level SQL guard **blocked 48/48 red-team attacks** (stacked queries, data-modifying CTEs, exfiltration, DoS) with **0/18 false positives**, covered by 92 tests.

CAF-OTSRNet: Optical-Guided Thermal Super-Resolution | [GitHub](#) | [Demo](#)

PyTorch, cross-attention, deformable alignment

- Developed a **cross-attention fusion network** that reconstructs high-resolution thermal imagery from low-resolution thermal and high-resolution optical satellite inputs, reaching **31 dB PSNR and 0.85 SSIM on Landsat-8**.

DataMind: LLM-Assisted Data Profiling | [GitHub](#)

Python, Ollama (local LLMs), Pandas, Streamlit, MySQL

- Built a local-LLM EDA platform that infers **foreign-key relationships via graph modelling** across multi-table datasets, flags anomalies in 7 categories, and proposes research directions.

ACHIEVEMENTS

- **First Prize**, DTU Summer School on AI (in collaboration with Adobe).
- **National Finalist, Smart India Hackathon 2025** — led the team on an ISRO-sponsored problem statement.
- **Top 53 of 2,000+ teams**, NatWest Code for Purpose Hackathon (FinSight: FastAPI forecasting + LLM scenario parsing).

SKILLS

Languages: Python, C/C++, Java, SQL

Agents & LLMs: Claude API tool use, MCP servers & clients, agent evaluation & benchmark design, LangGraph, RAG

ML: PyTorch, diffusion models, GANs, computer vision, experiment design & ablations

Systems: Docker, FastAPI, PostgreSQL, OAuth / JWT, AWS (S3, IAM, EC2), Linux, HPC / multi-GPU, Git, pytest

CS Core: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP